

General Processor Information

(Last Modified: November 7, 1995)

Processor		Source	Date (ship)	Bits (i/d)	Clock (MHz)	SPEC-92		SPEC-95		Units / Issue	Pipe Stages int/ldst/fp	Cache (i/d)		V _{dd} (V)	Tech (um)	M _e t	Power (W)		Size (mm ²)	Xsistor (10 ⁶)
						int	fp	int	fp			kB	Assoc				peak	typ		
Intel x86	i386SX	[41]	88	v/32	33	6.2	3.3			1/1	4/na/na	NA	NA	5.0	1.0	2	~2		43	0.28
	i486	[45]	89	v/32	33					2/1	5/na/5?	8 u.	4	5.0	1.0	2				1.2
	i486DX	[10]	91	v/32	50	27.9	13.1			2/1	5/na/5?	8 u.	4	5.0	0.8	3	5.0	3.9	81	1.2
	i486DX2	[20]	91	v/32	66	32.2	16.0			2/1	5/na/5?	8 u.	4	5.0	0.8	3	7.0	4.9	81	1.2
	i486DX4	[42]	93	v/32	99	51	27			2/1	5/na/5?	16 u.	4?	3.3	0.6	4		4		1.6
	P5	[2,10]	93	v/32	66	78	63.6			3/2	5/na/8	8/8	2/2	5.0	0.8	3	16.0		296	3.1
	P54VRT	[vi]	94	v/32	75	89.1	68.5			3/2	5/na/8			2.9	0.6	4	5.2	2.4		
		[vi]	94	v/32	90	110	84.4			3/2	5/na/8			2.9	0.6	4	6.5	3.0		
	P54C	[2,11]	94	v/32	100	122	93.2			3/2	5/na/8	8/8	2/2	3.3	0.6	4	5.0		163	3.1
	P54CQS	[47,vi]	95	v/32	120	172	108	3.7	2.8	3/2	5/na/8	8/8	2/2	3.3	0.35	4	10.0		163	3.1
	P54CS	[53,vi]	95	v/32	133	191	121	4.1	3.1	3/2	5/na/8	8/8	2/2	2.9	0.35	4				
	P55C	[31]	95	v/32	155					3/2	5/na/8				0.35	4				
	P6	[vi]	95	v/32	150	276	220	6.1	5.4	7/3 ⁴	14/14/16 ⁴	8/8 ^A		3.1	0.6	4	29.2	23.2	306	5.5
		[vi]	95	v/32	200	366	283	8.1	6.7	7/3 ⁴	14/14/16 ⁴	8/8 ^A		3.3	0.35	4	35	28.1	195 ^A	5.5
		[vi]	96	v/32	166	327	261	7.1	6.2	7/3 ⁴	14/14/16 ⁴	8/8 ^B		3.3	0.35	4	29.4	23.4	195 ^A	5.5
Cyril x86	5x86 (M1sc)	[vi]	95	v/32	100					2/1	6/6/?	16 u.	4	3.45	0.65	3	3.5	3	144	2.0
		[vi]	95	v/32	120					2/1	6/6/?	16 u.	4							2.0
	6x86 (M1)	[46]	95	v/32	100					3/2	7/7/?	16 u.	4	3.3	0.65	3	10		394	3.0
		[pn,vi]	95	v/32	120					3/2	7/7/?	16 u.	4	3.3	0.65	5			225	3.0
		[pn,vi]	96	v/32	>133					3/2	7/7/?	16 u.	4	3.3	0.5	5			169	3.0
Nexgen	Nx586	[42,43]	95	v/32	93					5/1 ⁴	7-9 ⁴	16/16		4.0	0.5	5	16	9	199	3.5
	Nx686	[vi]	95	v/32	180					7/4		16/32	2/2		0.35	5				~6
AMD	Am486	[pn]	95	v/32	120					2/1					0.5	3		~3		
	Am5x86	[vi]	95	v/32	133					2/1		16 u.			0.35	3				
	K5	[39,vi]	95	v/32	100	130				6/3	5/5/5?	16/8	4/4	3.3	0.35	3			161	4.3

Processor		Source	Date (ship)	Bits (i/d)	Clock (MHz)	SPEC-92		SPEC-95		Units / Issue	Pipe Stages int/ldst/fp	Cache (i/d)		V _{dd} (V)	Tech (um)	M _e t	Power (W)		Size (mm ²)	Xsistor (10 ⁶)
						int	fp	int	fp			kB	Assoc				peak	typ		
Mot 68k	68020	[45]	85	v/32	25	NA	NA			1/1	?/na/na	NA/.25	NA/1	5.0	1.5	2				
	68030	[41,45]		v/32	50		NA			1/1	3/na/na	.25/.25	1/1	5.0	1.2	2			55	0.27
	68040	[10,45]	89	v/32	25	21	15			2/1	6/na/6	4/4	4/4	5.0	0.8	3	6.0		164	1.2
	68060	[15,21]	93	v/32	50	~60 ²	~45 ²			3/2	8/na/8	8/8	4/4	3.3	0.5	3	3.9		198	2.4
Power PC	601	[15,25]	93	32	50	40	60			4/3	4/5/6	32 u.	8	3.6	0.6	4	9.1	6.5	121	2.8
	601+	[9]	94	32	100	105	125			4/3	4/5/6	32 u.	8	3.3	0.5	5	5.6	4.0	74	2.8
	602	[43]	95	32	66	40				4/2	4?/5?/6?	4/4	2/2	3.3	0.5	4		1.2	50	1.0
	603	[2,6,25]	94	32	80	75	85			4/2	4/5/6	8/8	2/2	3.3	0.5	4	3.0	2.2	85	1.6
	603e	[44]	95	32	100	120	105			4/2	4/5/6	16/16	4/4	3.3	0.5	4	3.5		98	2.6
	603ev	[mw]	96	32	166	165	150			4/2	4/5/6	16/16	4/4	2.5	0.35	5		2.5	81	2.6
	604	[7,8]	94	32	100	128 ⁸	120 ⁸			6/4-	4/5/6	16/16	4/4	3.3	0.5	4	13.0	9.0	196	3.6
		[53]	95	32	133	176 ⁸	157 ⁸	4.1	3.3	6/4-	4/5/6	16/16	4/4	3.3	0.5	4		14	196	3.6
	604e	[mw]	96	32	150	250									0.35					
	613	[mw]	97		166	230														
	614	[mw]	97		266	500														
	615 ⁹	[mw]	96	32	150	220(i) PPC														
						138(i) x86														
	620	[pn,43]	95	64	133	225	300			6/4-	4/5/6	32/32	8/8	3.3	0.5	4	30.0		311	6.9
	630	[mw]	97?	64?	600	600														
Alpha	21064	[1,4,23]	92	64	200	133	200			4/2-	7/7/10	8/8	1/1	3.3	0.75		30.0		234	1.68
	21064a	[18,23]	94	64	275	194	293	4.2	6.3	4/2-	7/7/10	16/16	1/1	3.3	0.5	4	33.0		164	2.8
		[vi]	95	64	300	220	300			4/2-	7/7/10	16/16	1/1	3.3	0.5	4				2.8
	21066a	[34,37]	94	64	233	94	110			4?/2-	7/7/10	8/8	1/1	3.3	0.5	4	23.0		161	1.75
	21164	[22,21]	94	64	300	341	513	7.3	12.2	4/4-	7/7/9	8/8 ¹	1/1	3.3	0.5	4	50.0		299	9.3
		[vi]	95	64	333	400	570			4/4-	7/7/9	8/8 ¹	1/1	3.3	0.5	4				9.3
	21164a	[vi]	96	64	>417	~500				4/4-	7/7/9	8/8 ¹	1/1	3.3	0.35					9.3

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						int	fp	int	fp			kB	Assoc				peak	typ		
MIPS	R3000	[12,20]	88?	32	40	27.9	35.8			1/1	5/na/na	na/na	na/na	5.0	1.2		4.0			
	R6000 ⁵	[20,32]	91	32	66.7	40.6	45.1			1/1	5?/na/na	na/na	na/na						5	
	R4000	[20,32]	92	64	100	59	61			2/1	8/na/10	8/8	1/1		1.0				213	1.1
	R4200	[2,5]	93?	64	80	55	30			1/1	5/na/5	16/8	1/1	3.3	0.64	3	1.8	1.5	78	1.4
	R4300i	[52]	95	64	100	60	45			1/1	5/na/5	16/8	1?/1?	3.3	0.35		1.8		45	1.7
	R4400	[10,17]	92	64	150	88	97			2/1?	8/na/10	16/16	1/1	3.3	0.6	3	15		186	2.3
		[37,38]	94	64	200	141	143	3.9	?	2/1?	8/na/10	16/16	1/1	3.3	0.5	2			148	2.2
		[53]	95	64	250	180	178			2/1?	8/na/10	16/16	1/1	3.3	0.35				108	~2.2
	R4600	[13,37]	94	64	150	110	83			2/1	5/na/5	16/16	2/2	3.3	0.64	3	4.6	3.0	77	1.85
	R4700	[37]	95	64	175	~130	~100			2/1	5/na/5	16/16			0.6	3			73	1.85
	R8000	[20,37]	94	64	75	109	311			6/4-	5/5/?	16/16	1/1	3.3	0.7	3			596 ³	3.43 ³
		[vi]	95	64	90	132	396			6/4-	5/5/?	16/16	1/1							
	R10000	[28,29]	95	64	200	>300	>600			5/5	5/6/7	32/32	2/2	3.3	0.5	4	~30		298	5.9
PA-RISC	PA 1.1	[32]		32	66	51	82			1/1	5/na/na	na/na	na/na		1.0				196	0.58
	HARP-1	[41]		32	120	70	120													
	7100	[3, 10]	92	32	100	80	150	3.2	4.1	2/2-	5/na/6	na/na	na/na	5.0	0.8	3	23.0		202	0.85
	7150	[24]	94?	32	125	136	201	4.0	4.6	2/2-	5/na/6?	na/na	na/na	5.0	0.8	3	30		196	0.85
	7100LC	[24]	94	32	100	102	137	2.9	3.5	3/2	5/na/6?	1/na	?/na	5.0	0.8	3	~10		196	0.8
	7200	[24]	94	32	140	~150	~250	4.9	8.8	3/2	5/na/6?	na/2	na/64	4.4	0.55	3	30		210	1.26
	7300LC	[vi]	96	32	~150	~200	~275	~5.5	~7.3			64/64			0.5	3				9
	8000	[33,48]	95	64	200	>360	>550	8.6	15	10/4-	7/9/9	na/na	na/na		0.5	4			345	3.9
SH Hitachi	II	[21]	93	16/32	28.7	~8	NA			1/1	5/5/na	4 u.	4	5	0.8	3	0.4			
	III (7708)	[21]	94	16/32	60	~35	NA			1/1	5/5/na	8 u.	4	3.3	0.5	3	0.6		44	0.8

not available yet.

Power: peak power numbers used when available, else estimated from average power. Easier to find peak, rather than average.

Units: Functional units NOT including “system” unit; **Issue:** Peak instructions issued per cycle; “-”: Indicates limitations on peak issue rate.

Cache: On-chip cache. **ETR:** Energy-Throughput Ratio = Watts/SPECave². Measure of Energy Efficiency. The lower, the better. For 1/ETR, the higher the better.

¹: The 21164 has an *on-chip* 96kB L2 cache.; The P6 has a 256kB L2 in the same MCM package (but xsistors not counted in total).

²: Estimated that x60 is 3 times the performance of the x40.

³: Combined total for 2 chips.

⁴: Data for the RISC-like core. (AMD: rops; NextGen: RISC86 instructions; Intel: uops)

⁵: This was an ECL implementation of the MIPS II ISA..

⁶: External (in an MCM package)

⁷: Single-chip CPU / Combined MCM package.

⁸: These are the highest SPEC numbers reported so far for the 604, which fall below the original estimates
(100MHz - 160/165; 133MHz - 200/200)

⁹: This chip reportedly combines a Pentium and PPC604 on one chip (multiple die?).

A: The P6 has a 256kB L2 (15.5 million xsistors) in the same MCM package (but xsistors not counted in total).

B: The P6 has a 512kB L2 (31 million xsistors) in the same MCM package (but xsistors not counted in total).

SOURCES (* = Can be found in the CPU Info Center)

- [1] = ISSCC92
 - [2] = ISSCC94
 - [3] = Hot Chips IV
 - [4]* = <http://netlib2.cs.utk.edu/performance/html/PDStop.html>
 - [5]* = Preliminary Product Sheet for R4200
 - [6] = to be published.
 - [7]* = Press Release 4/19/94: 100MHz PPC604
 - [8] = uP 4/94 (as reported by M. Horowitz)
 - [9]* = Press Release 3/30/94: 100MHz PPC601
 - [10] = Spectrum 12/93
 - [11] = to be published.
 - [12] = Courtesy M. Horowitz, SPEC Table
 - [13]* = Product Sheet for R4600
 - [15] = CompCon 93
 - [16] = CompCon 94
 - [17]* = Product sheet for R4400
 - [18]* = DEC Press Release.
 - [19] = CY7C601 Data Book
 - [20] = SPECtable from John Dimarco.
 - [21] = Hot Chips 94.
 - [22]* = DEC Press Release 9/7/94
 - [23]* = John DiMarco's SPEC Table.
 - [24]* = HP's Online server (<http://www.wsg.hp.com/wsg/Strategy/strategy.html>)
 - [25]* = PowerPC WhitePapers (http://www.austin.ibm.com/tech/p2ppc_tech.html)
 - [27] = from Dave Jaggar, chief architect of the ARM.
 - [28]* = MIPS Press release
 - [29] = 10/17/94 EE Times
 - [30] = 10/10/94 EE Times
 - [31]* = Computergram News
 - [32] = A Guide to RISC Microprocessors. Michael Slater, 1992.
 - [33]* = from comp.arch (http://infopad/~burd/gpp/announce/pa8000_overview)
 - [34] = 11/21/94 EE Times
 - [35] = Leslis Kohn, "Ultrasparc", Invited Speaker, Micro 27 Conf.
 - [36] = from comp.arch -- Alain Lachapelle (alainl@cam.org)
 - [37] = uP 12/26/94
 - [38]* = MIPS Home Page (<http://www.mips.com>)
 - [39] = uP 10/24/94
 - [40] = uP 1/24/94
 - [41]* = CHIPS and SYSTEMS SPEC Chart, by Gary Snow.
(http://infopad.eecs.berkeley.edu/~burd/gpp/summary/snow_survey)
 - [42]* = CHIPLIST 7.2 by Aad Offerman
(<http://einstein.et.tudelft.nl/~offerma/chiplist.html>)
 - [43] = ISSCC95
 - [44] = Motorola PPC Home Page:
(<http://www.mot.com/PowerPC/prodinfo.html>)
 - [45] = Advanced Microprocessors, Daniel Tabak, 1991
 - [46] = uP 2/16/94
 - [47]* = Intel Press Release
 - [48] = Compcon 95.
 - [49] = Electronic News 3/20/95
 - [52]* = Press Release 4/17/95
 - [53] = uP 6/19/95
 - [vi] = Vendor Info
 - [vp]* = Vendor Press Release
 - [pn]* = PowerPC News
 - [mw] = MacWeek (online)
- NOTE: Hobbit SPEC calculated from 28k Drystones $\approx$ 8 SPECint92
Hobbit = 27k Dry @ 20 MHz